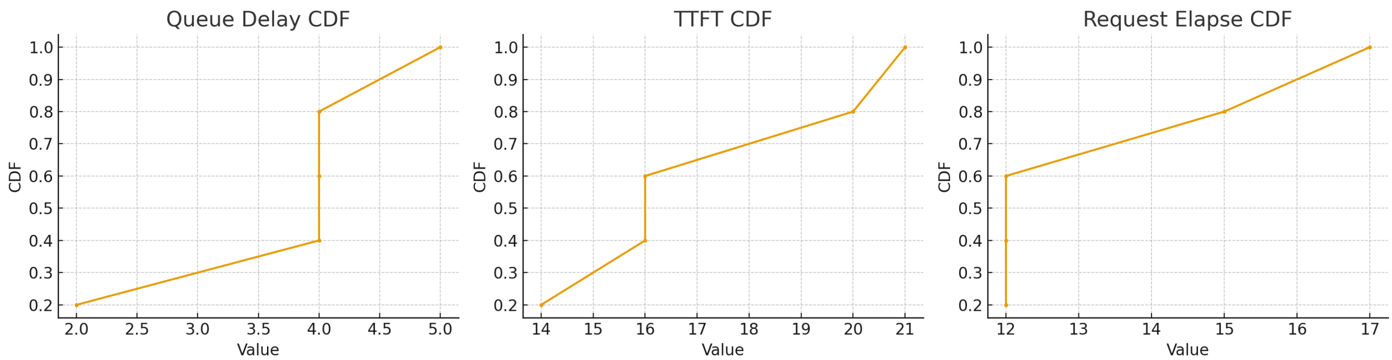
CDF calculation and plots

Help me calculate the CDF of the 3 items and draw 3 sub plots in a figure:

```
queue_delay = []
ttft = []
req_elapse = []

for req_id, (admitted, batching, finish) in req_id_tracker.items():
    queue_delay.append(batching - admitted)
    ttft.append(finish - admitted)
    req_elapse.append(finish - batching)
```



Here’s the figure with three subplots, each showing the CDF of your collected metrics:

- **Left:** Queue Delay
- **Middle:** Time-to-First-Token (TTFT)
- **Right:** Request Elapse

Each subplot plots the sorted values against their cumulative probability.

Do you want me to make the axes share the same y-axis scale for easier comparison?

I want the code

Got it — here's a clean version of the code you can use directly with your own `req_id_tracker`:

```
import numpy as np
import matplotlib.pyplot as plt

# Assume you already have req_id_tracker = {req_id: (admitted, batching, finish), ...}
queue_delay = []
ttft = []
req_elapse = []

for req_id, (admitted, batching, finish) in req_id_tracker.items():
    queue_delay.append(batching - admitted)
    ttft.append(finish - admitted)
    req_elapse.append(finish - batching)

def plot_cdf(data, ax, title):
    sorted_data = np.sort(data)
    yvals = np.arange(1, len(sorted_data) + 1) / len(sorted_data)
    ax.plot(sorted_data, yvals, marker=".", linestyle="-")
    ax.set_title(title)
    ax.set_xlabel("Value")
    ax.set_ylabel("CDF")
    ax.grid(True)

fig, axs = plt.subplots(1, 3, figsize=(15, 4), sharey=True)

plot_cdf(queue_delay, axs[0], "Queue Delay CDF")
plot_cdf(ttft, axs[1], "TTFT CDF")
plot_cdf(req_elapse, axs[2], "Request Elapse CDF")

plt.tight_layout()
plt.show()
```

This version shares the **y-axis** across all three subplots so you can compare the CDFs more easily.

Do you also want me to log-scale the x-axes? That's often helpful if delays have a heavy-tailed distribution.